

The Model Store a certified retrieval machine

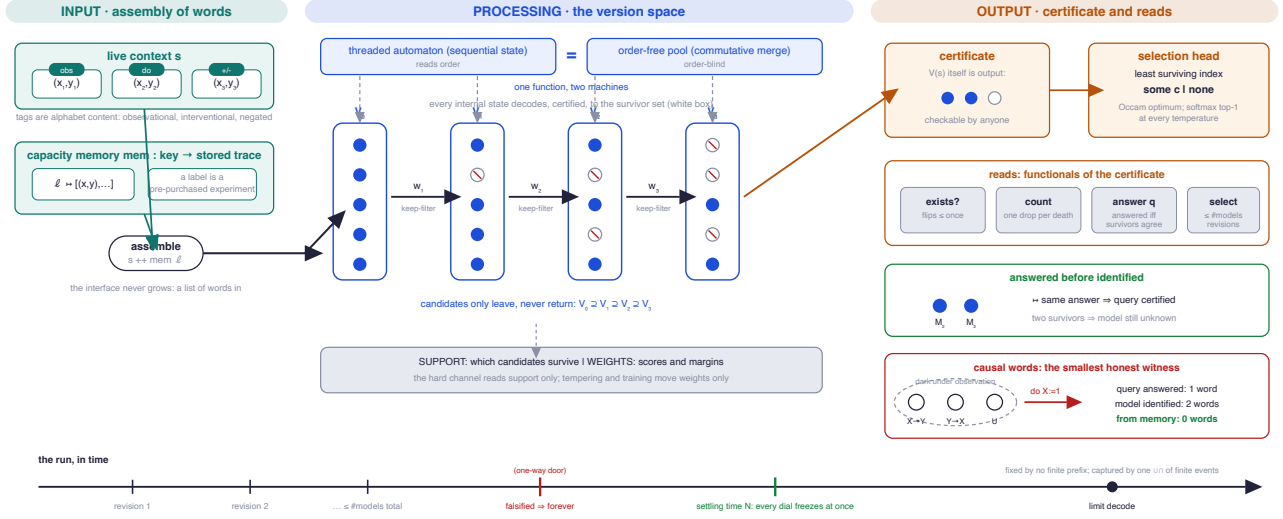
A constructed object. Every property below is a machine-checked theorem, exact rather than asymptotic; the stated instances execute inside the proof checker.

The object. A finite list of candidate models $c_1, \dots, c_k$, each a program $Dc : X \rightarrow Y$, operated as a retrieval machine over observation words.

INPUT. A context: a finite list of words (x, y) , assembled from live evidence and from memory (a label is a key whose stored trace is appended to the context). Tags on words carry observational, interventional, and negated readings. The interface is this one list type; it never grows.

PROCESSING. Maintain the *version space* $V(s)$: the candidates consistent with every word so far. One observation is one keep-filter; the space only shrinks. Computed by a sequential state machine or by an order-free commutative merge, which agree on every context. Every internal state decodes, with a checked certificate, to $V(s)$.

OUTPUT. The least-indexed survivor (or *none*), together with $V(s)$ itself as the certificate. All further answers are reads of the certificate: existence, counts, and per-model query functionals, each certified against every survivor.



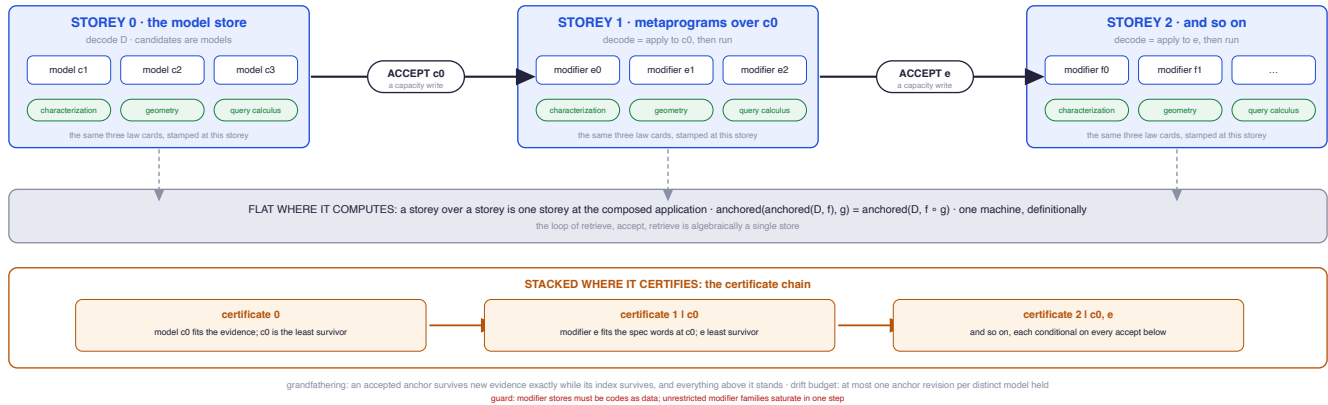
PROVEN PROPERTIES

- P1. One function, two machines.** The threaded state machine and the order-free merge compute the same answer on every context, for every store, with no side condition.
- P2. White box.** Every reachable internal state decodes to the survivor set, and any two faithful decodings agree along the run: there is no hidden internal freedom.
- P3. Exact capacity price.** The smallest machine reproducing the store's answers has size exactly the number of reachable survivor configurations. Sequential state count and parallel width times precision are the same price in two currencies, exchanged by complementation.
- P4. Order is a resource.** Order-free processing recovers everything determined by the survivor set and provably cannot recover the order of eliminations; a sequential pass recovers it exactly.
- P5. Exact revision budget.** The answer changes at most once per distinct model held, and an adversarial data order spends the whole budget. Duplicate encodings of one model never cost a revision.
- P6. Universal revision inheritance.** The survivor set itself changes at most k times, and every read of it inherits that budget. The existence read flips at most once: refutation is permanent, consistency forever provisional.
- P7. Temperature-free Occam selection.** The output is at once the least-indexed survivor, the optimum under every complexity-decreasing weighting, and the softmax top choice at every temperature. Hardening the soft regime moves only the weights, never the answer.
- P8. Dark knowledge located.** The weight stratum (scores, margins) is not a function of the survivor trajectory: two runs with identical survivor sets at every step can carry different weights. Everything the soft channel exposes beyond the answer lives in the losers.
- P9. Simultaneous settling.** On any unbounded stream the survivor set is eventually constant, and every readable dial freezes at one common time.
- P10. Identification at the limit.** If some candidate fits the whole stream, the limit answer is a candidate consistent with every prefix, reached within the revision budget. A target outside the list is approximated by the simplest consistent candidate, never identified: the machine's one boundary.
- P11. The limit, exactly bracketed.** No finite context determines the limit answer, yet the limit event is fully captured by one countable union of countable intersections of finite-context events.
- P12. Answered before identified.** A query is answered the moment all survivors agree on its value, with a certificate, possibly long before the model is pinned; answering can be strictly cheaper than identifying.
- P13. Labels are pre-purchased experiments.** A label's whole effect is the intersection law of its stored trace. It cannot separate candidates its stored words cannot kill, and a query can cost zero live words when the right experiment is already in memory.
- P14. Evidence of every polarity is alphabet content.** Demands to fail on a word, and words read under intervention, enter by recoding the alphabet; every property above transfers to the recoded store unchanged. On the smallest causal store, mechanisms indistinguishable under observation are separated by one interventional word: the query is answered at price one, the model identified at price two, and the same query costs zero from memory.

Kernel-checked in Lean 4 over Mathlib; every theorem's axiom footprint is the standard triple or less; the causal and exchange instances above are computed end-to-end by the checker's decision procedure. Selection, prices, and budgets are equalities. The weight stratum, distributional refinements, and the exact price law of experiments are stated obligations of the next construction, not properties claimed here.

The Second Storey metaprogram retrieval, and the limits of finiteness

The argument. A retrieved model can be accepted as the *anchor* of a next store, whose candidates are modifiers: codes that apply to the anchor and run. The anchored decode is a decode, so a store of modifiers is a store, and retrieval one storey up is the same machine. The query “here is my model, retrieve the modifier meeting my new conditions without touching the structure I accepted” is ordinary retrieval at the second storey; the loop of retrieve, accept, retrieve closes into a cascade.



- M1. Laws at every storey.** Every anchored store, for every anchor and every family of modifiers, carries the complete characterization, the full geometry, and the whole query calculus of pages one and three. The analysis of the machine is storey independent: one theorem covers all levels at once.
- M2. Flat where it computes, stacked where it certifies.** Composing anchorings is, definitionally, one anchoring at the composed application: the cascade is a single store at the level of computation. What stacks is the certificate chain: each acceptance freezes an anchor, and every downstream certificate is conditional on the accepts below it.
- M3. The chained certificate carries the freeze.** A base retrieval and an anchored retrieval certify jointly: the anchor fits its evidence, the modifier fits its evidence at that anchor, and each is its own stage's least survivor. The greedy selection is part of the certificate, not an unstated convention.
- M4. Grandfathering, with an exact drift budget.** An accepted

anchor survives new evidence exactly while its index survives, and while it does, everything built on it stands unchanged. The anchor itself is revised at most once per distinct model held: the model-grain revision law, read at the base store.

- M5. Correctness is evidence.** A modifier meets a specified behavior on given inputs exactly when it fits the words assembled from that specification. Checking a candidate surgery against its specification is a run of the machine on specification words, and the resulting certificate says so: agreement with the specification, never truth about the world.
- M6. Where finiteness ends.** At a countably infinite store, identification on realizable streams survives (the ordering of the indices replaces the finite count), and the per-word laws survive with it. What dies is exactly the uniform part: no store-size bound on revisions exists, unconditional settling fails off realizability, and selection is no longer computable, only specifiable. Finiteness is the price of the uniform, exact laws; it was never the price of identification.

Modifier stores obey the machine's own guard: candidates are codes as data (unrestricted modifier families collapse the storeys in one step). Retrieval over level-sorted modifiers returns the least-level fit; anchoring adds nothing to the selection price, and execution pays the two-level product. Adjacent frameworks (adapter fine-tuning, library learning, partial evaluation, knowledge compilation, specification monads) share the shape; none certifies its retrieval or prices it exactly.

The Witness, by Construction

one causal question through every storey

Every table cell and every survivor-set transition in the figure below is evaluated inside the proof checker. The figure does not illustrate the argument: it exhibits it. Reading the tables and checking each elimination by hand reproduces the proofs. Three rival mechanisms are indistinguishable under observation; one correctly chosen intervention answers the pressing question while two suspects still stand; the second intervention identifies the mechanism; the accepted mechanism anchors a store of surgeries from which the requested modifier is retrieved on specification words, with a joint, conditional certificate; and two mechanisms agreeing under every separate intervention are told apart only by the twin question that binds one hidden seed across two regimes.

